

Task No.	Task 1(B)
Title	Create a Webpage titled “My Styled Webpage”
Course Outcome	CO1

1. Aim

To design a webpage titled “My Styled Webpage” that uses the HTML5 <canvas> element to draw basic shapes and text (Part A), and CSS3 properties to style a header, navigation bar, content cards, and footer (Part B).

2. Objective

To understand and apply HTML5 Canvas drawing APIs (rectangle, circle, line, text) and core CSS3 styling properties — gradients, shadows, borders, spacing, and hover effects — to build one complete, visually styled webpage.

3. Theory

The HTML5 <canvas> element provides a drawable region controlled by JavaScript through the Canvas 2D Context API. Methods such as fillRect() draw rectangles, arc() combined with fill()/stroke() draws circles, moveTo()/lineTo()/stroke() draw lines, and fillText() renders text directly onto the canvas.

CSS3 extends basic styling with properties such as linear-gradient() for smooth colour transitions, box-shadow and text-shadow for depth and emphasis, border-radius for rounded corners, and the :hover pseudo-class for interactive feedback on links and buttons. Combined with margin, padding, and font properties, these allow a page to be structured (header, nav, content, footer) and styled consistently.

4. Algorithm

1. Start.
2. Create a valid HTML5 document and set the page <title> to “My Styled Webpage”.

Part A: HTML5 Canvas

1. Add a <canvas> element with a defined width and height.
2. Get its 2D drawing context using getContext("2d").
3. Draw a rectangle using fillRect().
4. Draw a circle using arc(), then fill() and stroke().
5. Draw a straight line using moveTo(),.lineTo(), and stroke().
6. Display the student's name as text using fillText().

Part B: CSS3 Styling

1. Create a header with a gradient background and text shadow.
2. Create a navigation bar with links and a hover effect.
3. Create three content cards with borders, border-radius, box-shadow, and padding.
4. Add a button inside each card with a hover effect.
5. Create a footer with a background colour and centered text.
6. Apply different font families and sizes across the header, cards, and navigation.

3. Save the file, open it in a browser, and verify the canvas drawing and all CSS3 effects render and respond correctly.
4. Stop.

5. Program (Source Code) — task1(B).html

task1(B).html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>My Styled Webpage</title>
<style>

  body {
    margin: 0;
    padding: 0;
    font-family: Arial, sans-serif;
  }

  /* Header */
  header {
    background: linear-gradient(to right, #4facfe, #00f2fe);
    color: white;
    text-align: center;
    padding: 25px;
    text-shadow: 5px 2px 4px rgba(0,0,0,0.4);
  }

  header h1 {
    font-size: 32px;
    font-family: 'Trebuchet MS', sans-serif;
  }

  /* Navigation Bar */
  nav {
    background-color: #333;
    padding: 12px;
    text-align: center;
  }

  nav a {
    color: white;
    text-decoration: none;
    margin: 0 15px;
    font-size: 16px;
```

```
font-family: Verdana, sans-serif;
}

nav a:hover {
  color: #00f2fe;
  text-shadow: 1px 1px 3px black;
}

/* Canvas Section */
.canvas-section {
  text-align: center;
  padding: 20px;
}

canvas {
  border: 1px solid #000;
  background-color: white;
}

/* Content Cards */
.card-container {
  display: flex;
  justify-content: center;
  gap: 20px;
  padding: 30px;
  flex-wrap: wrap;
}

.card {
  background-color: #f9f9f9;
  border: 1px solid #ccc;
  border-radius: 12px;
  padding: 20px;
  width: 220px;
  box-shadow: 4px 4px 12px rgba(0,0,0,0.2);
  text-align: center;
}

.card h3 {
  font-family: Georgia, serif;
  color: #333;
}

.card button {
  margin-top: 10px;
```

```
padding: 8px 16px;
border: none;
border-radius: 6px;
background-color: #4facfe;
color: white;
cursor: pointer;
}

.card button:hover {
background-color: #00c6ff;
box-shadow: 2px 2px 8px rgba(0,0,0,0.3);
}

/* Footer */
footer {
background-color: #333;
color: white;
text-align: center;
padding: 15px;
margin-top: 20px;
}

</style>
</head>
<body>

<!-- Header -->
<header>
  <h1>My Styled Webpage</h1>
</header>

<!-- Navigation Bar -->
<nav>
  <a href="#">Home</a>
  <a href="#">About</a>
  <a href="#">Services</a>
  <a href="#">Contact</a>
</nav>

<!-- ===== PART A: HTML5 Canvas ===== -->
<div class="canvas-section">
  <h2>Part A - HTML5 Canvas</h2>
  <canvas id="myCanvas" width="400" height="300"></canvas>
</div>
```

```
<script>
  const canvas = document.getElementById("myCanvas");
  const ctx = canvas.getContext("2d");

  // 1. Draw a Rectangle
  ctx.fillStyle = "skyblue";
  ctx.fillRect(30, 30, 120, 80);

  // 2. Draw a Circle
  ctx.beginPath();
  ctx.arc(280, 70, 50, 0, 2 * Math.PI);
  ctx.fillStyle = "orange";
  ctx.fill();
  ctx.stroke();

  // 3. Draw a Line
  ctx.beginPath();
  ctx.moveTo(30, 180);
  ctx.lineTo(350, 180);
  ctx.strokeStyle = "black";
  ctx.lineWidth = 3;
  ctx.stroke();

  // 4. Display Name as Text
  ctx.font = "24px Arial";
  ctx.fillStyle = "darkblue";
  ctx.fillText("Akshaya", 130, 250);
</script>

<!-- ===== PART B: CSS3 Styling ===== -->
<h2 style="text-align:center;">Part B - CSS3 Styling</h2>

<div class="card-container">
  <div class="card">
    <h3>Card One</h3>
    <p>This is the first content card with styling.</p>
    <button>Learn More</button>
  </div>

  <div class="card">
    <h3>Card Two</h3>
    <p>This is the second content card with styling.</p>
    <button>Learn More</button>
  </div>
</div>
```

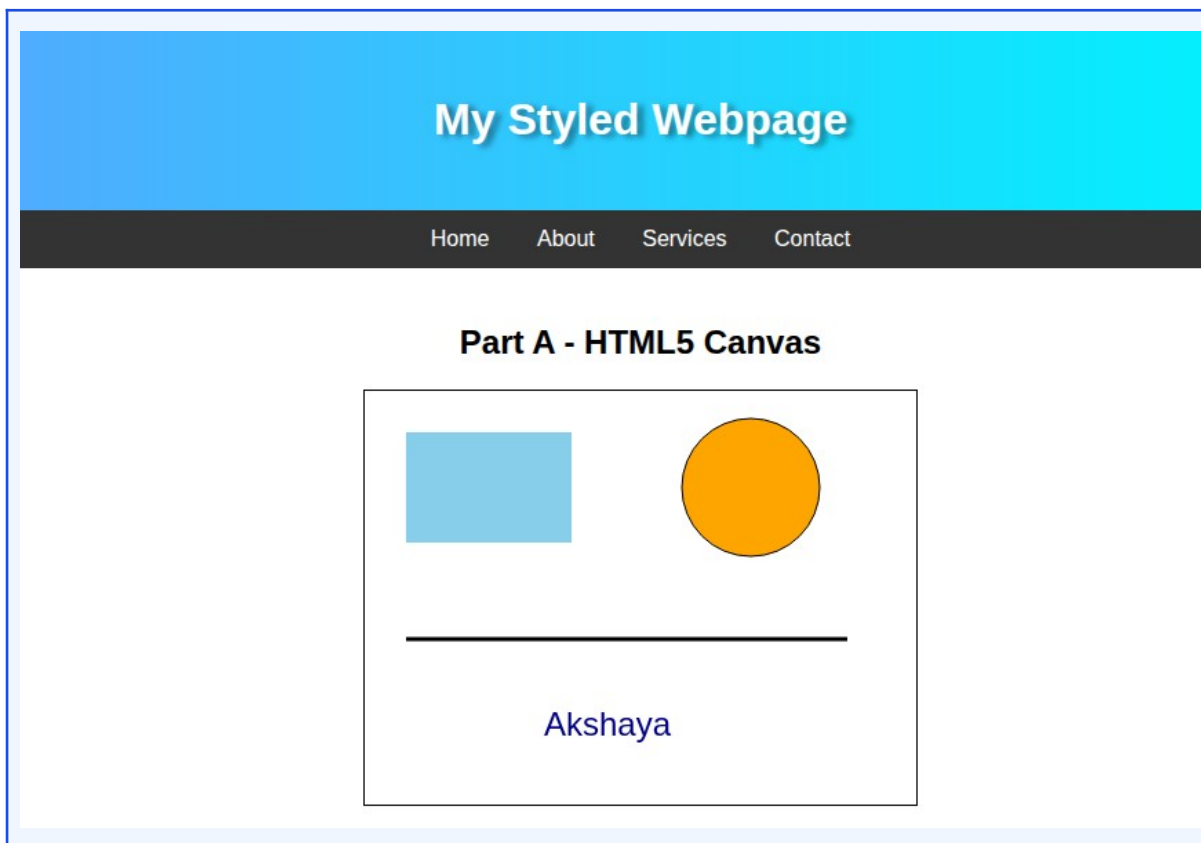
```
<div class="card">
  <h3>Card Three</h3>
  <p>This is the third content card with styling.</p>
  <button>Learn More</button>
</div>
</div>

<!-- Footer -->
<footer>
  <p>&copy; 2026 My Styled Webpage. All Rights Reserved.</p>
</footer>

</body>
</html>
```

6. Expected Output

The screenshots below show task1(B).html rendered in a Chromium browser (in two parts, top to bottom), demonstrating the gradient header with text shadow, the navigation bar, the Part A canvas drawing (rectangle, circle, line, and name text), and the Part B content cards with border-radius, box-shadow, and hover-ready buttons, followed by the footer.



Part B - CSS3 Styling

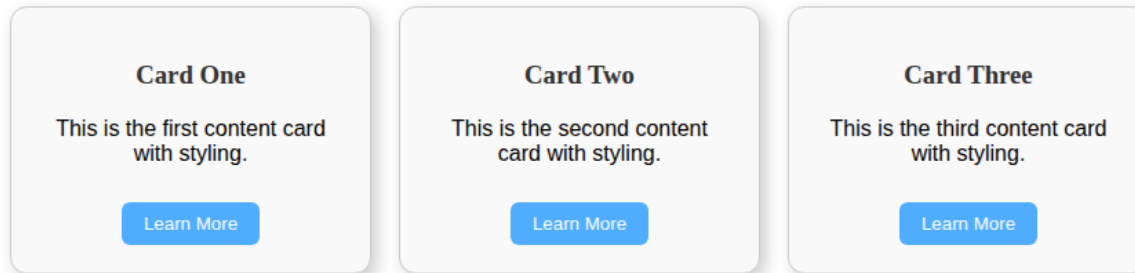


Figure 1: My Styled Webpage — rendered in Chromium browser (top to bottom).

7. Result

The webpage “My Styled Webpage” was successfully designed and executed using HTML5 and CSS3. The HTML5 canvas correctly drew a rectangle, a circle, a line, and the student's name as text, and the page was fully styled with a gradient header, text shadow, navigation bar with hover effects, three content cards with border-radius and box-shadow, hover-responsive buttons, and a styled footer, fulfilling Course Outcome CO1.

Student Signature

Faculty / Lab In-charge Signature